Quality — Reject Disposition Policy

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> **Audience:** Quality Worker · Line Manager · Plant Manager
> **Applies to:** Every part flagged `rejected` in the simulator's part event stream, on any of the 30 stations.

1. Purpose

This policy defines what happens to a part once it is flagged as a reject by any station's inspection or test. The decision tree is the same across all 5 lines but the **threshold values and dispositions differ by line and by station**.

2. Reject thresholds by station

Reject probability targets from the simulator and the operational limits above which the line should pause for triage:

Line	Station	Target reject rate	Pause threshold (1 hr rolling)
Line-A	CNC-Lathe	2 %	> 4 %
Line-A	CNC-Mill	3 %	> 5 %
Line-A	Surface-Grinder	2 %	> 4 %
Line-A	Deburring-Station	1 %	> 3 %
Line-A	CMM-Inspection	5 %	> 8 %
Line-B	Blanking-Press	2 %	> 4 %
Line-B	Hydraulic-Press	4 %	> 6 %
Line-B	Trimming-Station	2 %	> 4 %
Line-B	Quality-Inspection	3 %	> 5 %
Line-C	Welding-Robot	3 %	> 5 %
Line-C	Weld-Inspection	4 %	> 7 %
Line-	Fastening-		

C	Station	1 %	> 3 %
Line-C	Assembly-Robot	2 %	> 4 %
Line-C	Leak-Test	3 %	> 5 %
Line-D	Surface-Prep	1 %	> 3 %
Line-D	Chemical-Wash	1 %	> 3 %
Line-D	Primer-Application	3 %	> 5 %
Line-D	Paint-Booth	4 %	> 6 %
Line-D	Curing-Oven	2 %	> 4 %
Line-D	Coating-Inspection	3 %	> 5 %
Line-E	SMT-Placement	3 %	> 5 %
Line-E	Reflow-Oven	2 %	> 4 %
Line-E	AOI-Inspection	4 %	> 7 %
Line-E	Through-Hole-Insert	2 %	> 4 %
Line-E	Wave-Solder	3 %	> 5 %
Line-E	Functional-Test	5 %	> 8 %
Line-E	Conformal-Coat	2 %	> 4 %

When a line crosses any pause threshold, the **Line Manager pauses the station for triage**, and the relevant station SOP is consulted to identify the likely fault even if `machine_status` is still `Running`.

3. Disposition decision tree

For every rejected part:

1. **Capture context** — part ID, line, station, reject reason code, `fault_type` of any open work order on that station.
2. **Classify the defect**:
   - **Cosmetic** (surface finish, edge break, minor visual) → see step 4.
   - **Dimensional / functional** (outside tolerance, leak, electrical) → see step 5.
   - **Safety-critical** (e.g., weld penetration failure, leak on a pressure boundary) → see step 6.
3. **Apply line-specific routing** below.

4. Cosmetic defect

- **Rework** at the originating station if rework is technically feasible (e.g., Deburring re-passes, Coating-Inspection paint touch-up).
- If rework is not feasible, route to **scrap** with disposition code `COS-SCRAP`.

5. Dimensional / functional defect

- **Quarantine** the part until a Quality Engineer reviews.
- For Line-A CMM-Inspection rejects → check the prior 5 parts on CMM (sampling).
- For Line-C Weld-Inspection or Leak-Test rejects → hold the **entire buffer of parts produced after the last good cycle**.
- For Line-E AOI-Inspection or Functional-Test rejects → hold the individual board; route to engineering rework if cost-justified, else scrap with code `FUNC-SCRAP`.

6. Safety-critical defect

- **Immediate quarantine** of the part with red tag.
- Notify Line Manager and Quality Engineer.
- Trigger a **3-piece holdback rule** on the affected station: hold the prior 3 parts in addition to the rejected one until QE has reviewed.
- Create a maintenance work order against the station even if `machine_status` is still `Running` — this often indicates a brewing fault not yet escalated by telemetry.

7. RMA & customer returns

Returned product from the field is processed by Quality Engineering and not covered by line dispositions. RMA disposition codes (`RMA-INVESTIGATE`, `RMA-CREDIT`, `RMA-REPAIR-RETURN`) are tracked separately.

8. References

- Each station SOP §3 (“Idle reasons”) and §4 (“Faults & corrective actions”).
- `Maintenance_Workflow_Overview.pdf` for the work-order lifecycle.
- `OEE_Targets_and_Escalation.pdf` for the Quality component of OEE.
- `Production_Lines_Reference.pdf` for line topology.